

Two-Phase State Justification Recovers Partial-Scan ATPG Coverage

A progressive residual study on timing-constrained
partial-scan circuits

VLSI Testing — Group 5

丁睿濂 駱彥竹 黃思維

June 2026

Partial Scan Creates an X-State Barrier

- Scan-MUX delay breaks timing → top 10% FFs kept non-scan
- At one frame, non-scan FF outputs are unknown (X)



X on the path ⇒ fault effect cannot propagate ⇒ structurally unstable

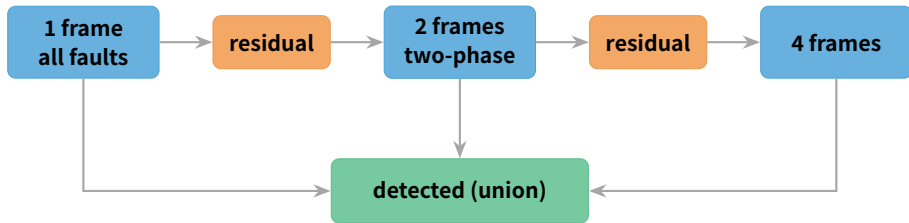
Key Finding

Most lost partial-scan coverage is **not** about
insufficient **depth**, search **budget**, or **heuristics**
it is **interleaved propagation & state justification**

Two-phase decoupling recovers most of the full-scan gap:

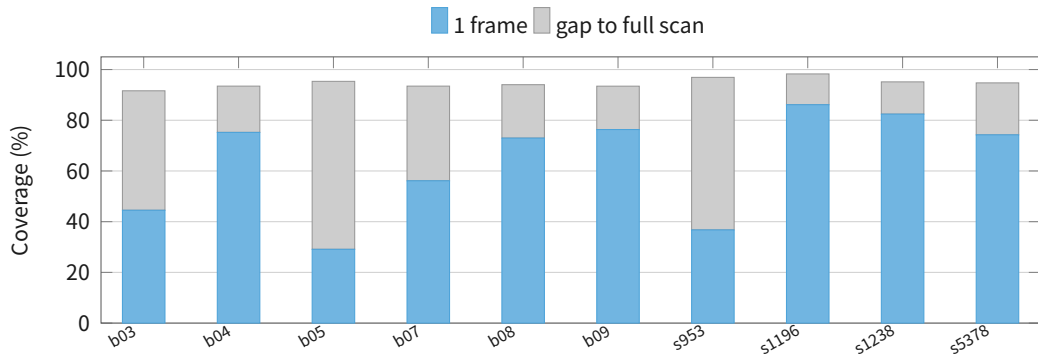
31.26 pp $\rightarrow$ **4.24 pp**

Method & Setup



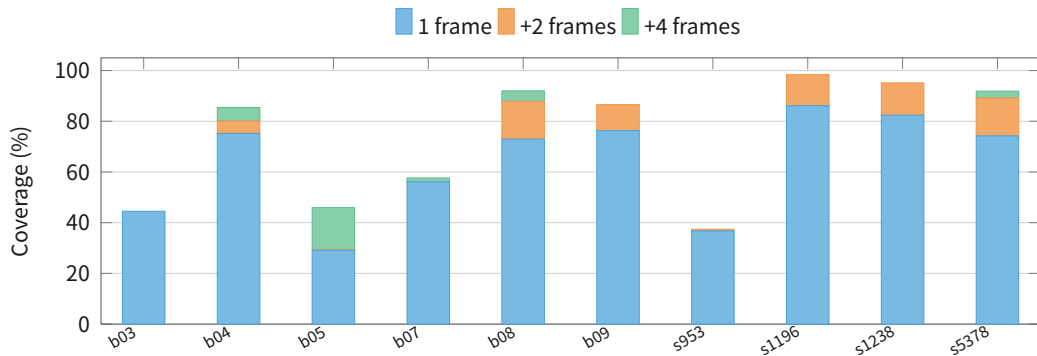
- 10 circuits: ITC'99 + ISCAS'89
- Collapsed stuck-at; one ATPG engine
- Non-scan = OpenSTA slack, bottom 10%
- Each stage targets only the residual
- Backtrack 800 / 5000; no timeout
- Full-scan baseline: same engine & limits

Step 1 — The Gap to Full Scan



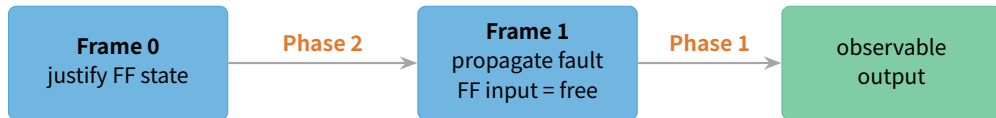
Bar top = full scan; the **grey** part is the gap — average **31.26 pp** (up to 66)

Step 2 — Baseline Recovery is Inconsistent



Two-phase OFF: +10.10 pp avg — but **b05** gains at 4 frames, **s1196** at 2. Why?

Hypothesis B: Two-Phase State Justification



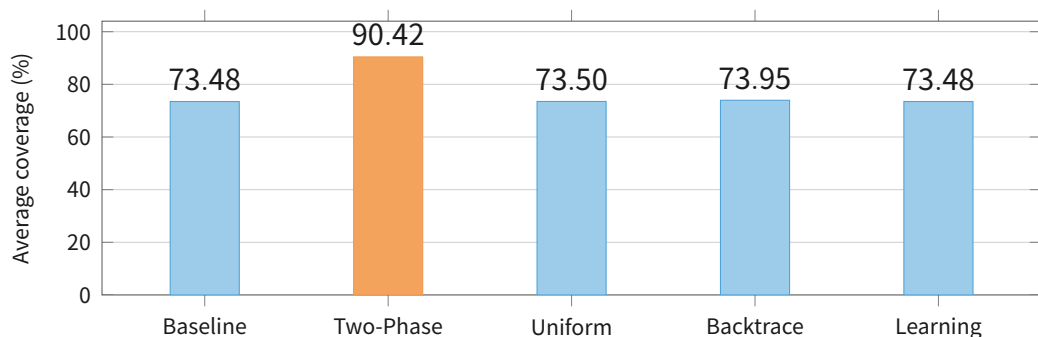
- **Phase 1** (last frame): FF inputs **free** → PODEM finds a path
- **Phase 2** (earlier frames): **justify** those values backward
- Propagation no longer **blocked by X**, no budget competition

The Five Controlled Experiments

#	Change vs baseline	Isolates
1	— (baseline)	plain pipeline
2	Two-phase ON	propagation / justify decoupling
3	Uniform 5000 at 1 frame	backtrack-limit differential
4	Enhanced backtrack	backtrace heuristic
5	Static learning	early conflict detection

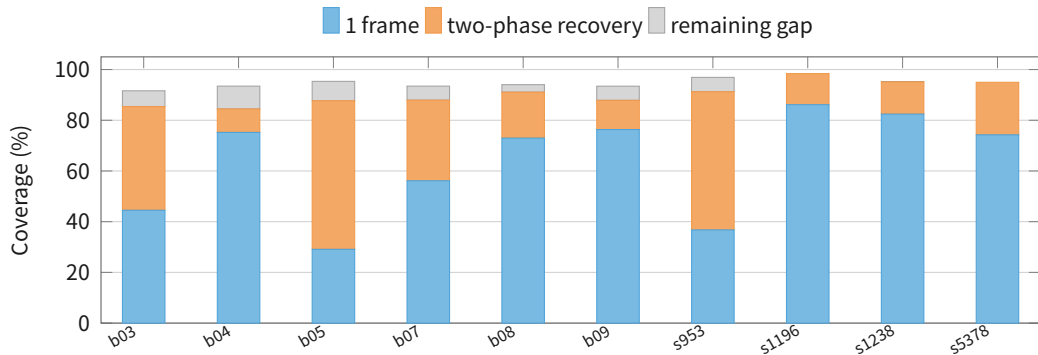
Same 10 circuits, full 1→2→4 pipeline, one parameter changed each

Ablation Result — Only One Lever Moves



Two-phase ON: +16.94 pp. Uniform limit +0.02, backtrace +0.47, learning +0.00 pp

Two-Phase Closes the Gap



Average gap to full scan: 31.26 pp → 4.24 pp (s1196 / s5378 even exceed it)

Conclusion

- Two-phase state justification is the dominant mechanism (+16.94 pp)
- Backtrack limit, backtrace, learning: negligible (+0.02 / +0.47 / +0.00 pp)
- Gap to full scan: 31.26 pp $\rightarrow$ 4.24 pp
- 4 frames redundant once the 2-frame search is structured right

The bottleneck is search structure — not budget, depth, or heuristics

Thank You

Questions & Discussion

Two-Phase Partial-Scan ATPG · VLSI Testing · Group 5

Backup Slides

Supporting detail for Q & A

Two Hypotheses

- **A. Backtrack-limit differential** (800→5000): residual exists only because 1 frame is capped
- **B. Two-phase justification:** decouple propagation from state justification
- Plus two controls: **enhanced backtrace, static learning**

Test each by changing **exactly one parameter** — a controlled ablation

Two-Phase: Why It Works (Walkthrough)

Single search (standard)

- Goal: propagate fault to an output
- Path needs non-scan $FF = 1$
- But FF is currently **X**
- Search does propagation + justification at once
- $\Rightarrow$ conflict $\rightarrow$ **abort**

Two-phase

- **Frame 1:** FF input free $\rightarrow$ propagate fault first
- **Frame 0:** justify that $FF = 1$ backward
- $\Rightarrow$ the two goals are **separated**
- $\Rightarrow$ each solvable $\rightarrow$ **detected**

Same fault, same depth — only the **search order** changed

Per-Circuit Recovery

Circuit	Baseline	Two-phase	Δ
b03	44.50%	85.41%	+40.91
b05	45.94%	87.68%	+41.74
s953	37.40%	91.26%	+53.86
b04	85.42%	84.46%	-0.96

Near-failure $\rightarrow$ near-full recovery; a few already-favorable circuits barely move

Why Four Frames Add Nothing

- Two-phase ON: 4 frames add 0 pp on every circuit
- 2 frames already catch all recoverable faults
- Without two-phase, 4-frame gain is just 2-frame inefficiency
- Residual after 2 frames = untestable at any depth

Frame depth is not the bottleneck — search structure is

Threats to Validity & Next Steps

Threat	Current	Next step
Non-scan ratio	only 10%	sweep 5 / 10 / 20 / 30%
Engine dependence	one ATPG engine	cross-check another engine
Depth claim	up to 4 frames	test 8 frames selectively
Excluded circuits	5 excluded	debug crash / timeout

Also bounded by the 800 / 5000 limit choice and operational “untestable”